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Numerical Simulation of Acid-Rock Dissolution in Porous Carbonate Reservoirs Under Different Injection Rates

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Summary

Calcite dissolution plays a vital role in geological, hydrological, and environmental engineering, particularly in carbon sequestration, rock weathering, and groundwater chemical evolution. When acidic fluids react with calcium carbonate minerals like calcite, they often form typical wormhole structures. Understanding the morphological evolution of dissolution channels like wormholes in carbonate rocks under acid injection is critical for optimizing applications such as acid stimulation and geological carbon sequestration. As a key factor in wormhole formation, injection rate governs whether the dissolution reaction develops into a wormhole structure either dispersed or planar-front. In this study, the widely used open-source package PorousMedia4Foam is employed to simulate fluid flow and acid-rock dissolution transport in the free flow region of porous carbonate reservoirs, taking into account the effects of acid injection rate on the formation of wormholes in both homogeneous and structurally heterogeneous porous media, as well as the formation and temporal evolution of rock wormhole structures. Results show that in homogeneous media, wormhole formation is most efficient at intermediate velocities, where reaction and transport are balanced. However, in heterogeneous media, acid fronts are significantly perturbed by structural irregularities, leading to increased branching, tortuous paths, and the loss of well-defined wormhole morphology. These findings highlight the necessity of incorporating geological heterogeneity into predictive models of reactive transport, particularly in optimizing acidizing treatments and assessing storage integrity in subsurface formations. In summary, this study provides a theoretical foundation for a deeper understanding of wormhole evolution during reactive seepage and offers valuable insights for engineering applications such as oil and gas reservoir stimulation and geological carbon sequestration.

Key Words: Calcite dissolution, Wormhole structures, Acid injection rate, PorousMedia4Foam, Carbonate reservoirs

Introduction

Calcium carbonate ($CaCO_3$), a mineral widely found in nature, plays a significant role in geology and hydrology. The solubility and reactive behavior of $CaCO_3$ under different environmental conditions make it a research hotspot in subsurface engineering and geochemical modeling. Acid-rock dissolution [1] is a widely used stimulation technique in the development of carbonate reservoirs. Its primary objective is to enhance oil and gas production by increasing reservoir permeability through chemical reactions between injected acid and carbonate rock [2]. Additionally, this technique can effectively create or enlarge seepage channels, improve the recoverability of low-permeability and tight carbonate formations, and reduce overall production cost.

In porous carbonate reservoirs, the reaction between acidic fluid and calcite often induces significant pore structure evolution, leading to the formation of wormhole with highly heterogeneous features. These wormholes are highly conductive, finger-like channels that significantly enhance fluid transport within low-permeability formations [3]. The formation of wormhole structures is a highly nonlinear and dynamic process [4, 5], governed by the complex interplay of multiple factors such as chemical reaction kinetics, mass transport, fluid dynamics, and the inherent heterogeneity of the porous medium. Among these, the dissolution reaction is closely coupled with fluid transport, with injection rate being a critical parameter that regulates this coupling. Specifically, injection rate not only affects the transport efficiency of reactive species but also alters the local concentration boundary layer. Overall, calcite, the primary crystalline form of calcium carbonate [6], exhibits dissolution kinetics that are highly sensitive to variations in acid injection rate. This flow-dependent sensitivity causes the distinct dissolution regimes, including uniform dissolution, face dissolution, and wormholing, each characterized by unique spatial patterns and varying effectiveness in enhancing reservoir permeability [7, 8]. Therefore, investigating the acid dissolution behavior of calcite in porous media under varying acid injection rates is of great significance for understanding mineral reaction mechanisms in natural systems and for improving predictive simulations in engineering applications.

Numerical simulation offers a powerful tool for studying the evolution of such dissolution features, enabling systematic analysis of the underlying mechanisms under controlled conditions. OpenFOAM, an open-source computational fluid dynamics (CFD) platform, is widely used for modeling complex multiphysics flow and transport problems. Here, the PorousMedia4Foam [9] module offers specialized capabilities for simulating reactive transport in porous media, making it particularly well-suited for investigating acid-rock interactions. Currently, research utilizing the PorousMedia4Foam module is primarily focused on the following areas: (1) simulation of reactive transport processes at the pore scale, with particular emphasis on accurately capturing acid–calcite interface reactions [10]; (2) investigation of the fundamental mechanisms driving wormhole formation [11]; (3) development of multiscale coupled simulation frameworks to bridge pore- and core-scale phenomena [12]; and (4) analysis of the influence of mineralogical heterogeneity on dissolution patterns [13]. In details, the correlation between porosity-permeability evolution can be well investigated, as well as the effects of some key parameters such as temperature and injection rate on dissolution kinetics and wormhole morphology [14, 15]. Furthermore, the multiphase flow model can be integrated with chemical reaction modules to simulate the dissolution behavior of calcite in carbonate rocks under varying acid concentrations [16]. Additionally, artificial intelligence (AI) techniques, such as machine learning, can be used to enhance the simulation accuracy and reduce calculation time, such as accelerating wormhole path prediction and enabling more efficient generation of wormhole structures [17, 18].

In summary, the reactive dissolution process of acid and calcite has an important impact on the formation of fluid flow channels during carbonate acidification and geological carbon sequestration. To gain a deeper understanding of the pore structure evolution induced by acid-calcite interactions, this study develops a numerical model of reactive porous media seepage based on the PorousMedia4Foam module within the open-source CFD platform OpenFOAM. The model enables simulation of acid transport and reaction dissolution at the pore scale, and accurately captures the interfacial reaction dynamics at the acid-calcite interface. By varying the acid injection rate, the formation and evolution of wormholes under different flow conditions were systematically investigated. Simulation results indicate that flow velocity is a critical factor influencing the morphology and breakthrough efficiency of wormholes. At low injection rate, dissolution tends to occur uniformly across the medium, and at moderate injection rates, highly channelized and directional wormhole structures are more likely to develop. However, high injection rate may lead to a decline in overall dissolution efficiency. This study not only elucidates the pore-scale mechanisms underlying acid-rock reaction kinetics but also provides a theoretical reference for optimizing operational parameters in acidizing treatments. Fundamentally, systematically investigating acid injection rate and its effects on calcite dissolution advances theoretical geochemistry while supporting the development of more effective engineering strategies for resource extraction and environmental management.

Model developments

A generic platform, **PorousMedia4Foam** [19], developed within the OpenFOAM framework, is employed to construct a two-dimensional coupled reaction-flow-transport model. This model is used to solve flow and transport in porous media at various scales, and to investigate wormhole evolution induced by the reaction between acidic fluid and calcite under varying flow rates. The simulation domain consists of a porous medium with a uniform initial porosity. By varying the acid injection rate, the characteristics of wormhole formation, as well as the evolution of porosity and reaction rate, are systematically analyzed. PorousMedia4Foam uses the Darcy-Brinkman-Stokes (DBS) equation to solve the flow and transport in solid-free and porous regions within a single framework [20]. The DBS equation is derived from the integration of the Navier-Stokes equations over a control volume [21]. When the porosity equals 1, the Navier-Stokes equations is solved. However, when the porosity is between 0 and 1, the Darcy equations is solved [20]. The DBS equation effectively describes fluid flow and species transport, offering a robust framework for capturing the impact of wormhole evolution during the acid-induced dissolution of carbonate rocks. It can well describe the momentum transport of acid fluid throughout the whole region during the acidizing process, avoiding the definition of complex interfacial conditions between the porous media and the free-flow region through experimental or numerical methods.

Flow solver and equation

In this study, PorousMedia4Foam employs the multi-scale flow solver dbsFoam [22], which is based on the micro-continuum approach, to simulate acid-rock dissolution processes in carbonate reservoirs. dbsFoam, short for Darcy-Brinkman-Stokes Foam, is a specialized solver within PorousMedia4Foam designed to model reactive fluid flow and mass transport in porous media. This solver integrates porous media flow models based on the Darcy, Brinkman, and Stokes formulations [21], and supports complex processes such as multi-component transport, chemical reactions, and porosity evolution. It is particularly well-suited for simulating reactive seepage phenomena, such as calcite dissolution and wormhole formation. The micro-continuum method relies on the Darcy-Brinkman-Stokes (DBS) equations [23, 24], which allow the simulation of flow and transport in both solid-free and porous regions within a single framework [22]. The momentum equation is,

$$\frac{1}{\varphi}\left(\frac{\partial \rho_f v_f}{\partial t} + \nabla \cdot \left(\frac{\rho_f}{\varphi} v_f v_f\right)\right) = -\nabla p_f + \rho_f g + \nabla \cdot \left(\frac{\mu_f}{\varphi}\left(\nabla v_f + {}^t\nabla v_f\right)\right) - \mu_f k^{-1} v_f \tag{1}$$

Where φ is the porosity, v_f is the fluid flow velocity, p_f is the acid fluid pressure, g is the gravity, ρ_f is the fluid density, μ_f is the fluid viscosity and k are the cell permeability. Eq. (1) is valid throughout the entire computational domain. When φ = 1, the region consists purely of fluid, and the drag force $\mu_f k^{-1} v_f$ becomes zero, reducing the momentum equation to the Navier-Stokes form. On the other hand, when $0 < \varphi < 1$, the region comprises both fluid and solid phases, so Eq. (1) approximates Darcy's law because the drag force dominates over both inertial and viscous forces.

The impleFirstOrderKineticMole [25] is used to solve the reaction of acid-rock dissolution reaction. It serves as a simple geochemical engine that handles multiphase transport by applying a first-order kinetic reaction between a fluid-phase species and solid minerals. The mass balance equation for fluid-phase species labelled A is,

$$\frac{\partial \varphi C_A}{\partial t} + \nabla \cdot \left(v_f C_A\right) - \nabla \cdot \left(\varphi D_j \cdot \nabla C_A\right) = -\left(\sum_{j=1}^{N_S} A_{s,j}\left(k_{j,A}\gamma_A\right)\right) C_A \tag{2}$$

Where C_A is the species concentration, v_f is the fluid flow velocity, D_j is the diffusion tensor, $A_{s,j}$ is the reactive area of minerals j, d $(k_{j,A}\gamma_A)$ is the reaction constant between species A and minerals.

In impleFirstOrderKineticMole, the finite volume method is used to discretize the computational grid based on Eq. (2), and the equation is solved implicitly. In short, impleFirstOrderKineticMole is a model for describing the reaction rate R,

$$R = \left(k_{j,A}\gamma_A\right) \cdot C_A \tag{3}$$

The distribution of solid minerals j is,

$$\frac{\partial Y_{s,j}}{\partial t} = -A_{s,A}\left(k_{j,A}\gamma_A\right) V_{m_{s,j}} C_A \tag{4}$$

Where $V_{m_{s,j}}$ is the molar volume of minerals j. This reaction model is simple but efficient to investigate the mass transport through a single reaction. It assumes a first-order reaction, where the reaction rate is proportional to the concentration of a single reactant. The rate is solely controlled by the reactant concentration, without considering reverse reactions, and it does not depend on factors such as temperature, pH, or reactive surface area.

Porous media models

The acid-rock dissolution simulation is conducted within a porous media domain, where the performance of flow solvers and geochemical modules is strongly influenced by key porous media properties, including absolute permeability, specific surface area, and the dispersion tensor. These properties characterize pore-scale effects linked to the microstructure of the porous medium and can vary as the microstructure evolves during the acid-rock reaction. These porous properties are described in detail below.

Permeability Kozeny-Carman model. Absolute permeability describes the ability of porous media to conduct fluid flow and is inherently determined by the microstructure, and it changes with chemical processes such as mineral dissolution. In this study, the Kozeny-Carman model [26] is employed to represent the changes in permeability during the acid-rock dissolution process. This model effectively captures the direct influence of pore structure evolution on permeability, making it well-suited for dynamically describing changes induced by acid-rock dissolution reaction [27].

$$k = k_0\left(\frac{\varphi}{\varphi_0}\right)^n\left(\frac{1-\varphi_0}{1-\varphi}\right)^m \tag{5}$$

Where k is current permeability, k_0 is initial permeability, φ is current porosity, and φ_0 is initial porosity. The Kozeny-Carman (K-C) model is a classic semi-empirical formula that describes the relationship between permeability and pore structure evolution in porous media. In the simulation of acid-rock dissolution reactions, pore structure changes, such as pore expansion and increased connectivity, occur as the chemical reactions. The K-C model is widely used for permeability prediction from micro to macro scales, owing to its simple mathematical form and its ability to incorporate porosity variations. Moreover, the model has been extensively validated in acid-rock dissolution experiments involving various minerals, such as limestone, dolomite, and sandstone, and is frequently employed in carbonate dissolution simulations due to its demonstrated accuracy.

Surface area model. It is crucial to accurate estimate the reaction surface area for describing the chemical kinetics of dissolution in acid-rock dissolution. However, the reactive surface area is difficult to measure directly, as only a fraction of the total surface area typically participates in the chemical reaction. In the transport process dominated by advection, the reaction mainly occurs near the high-speed streamline [28], resulting in the actual reaction surface area being significantly smaller than the theoretical geometric area. In addition, as the acid continues to dissolve minerals, the mineral volume fraction gradually changes. The continuous evolution of the pore structure leads to a non-monotonic change in the specific surface area [29]. In details, the specific surface area increases in the early stage of wormhole formation due to dissolution. However, as the channels gradually expand and connect, the solid mineral fraction is reduced, leading to a decrease in specific surface area. In order to more accurately describe the relationship between specific surface area and mineral dissolution evolution, the Power-Law model is used here,

$$A_s = A_0 \left(Y_s\right)^n \tag{6}$$

Where A_s is current specific surface area, A_0 is initial specific surface area, Y_s is structure factor, and n is an empirical parameter. The Power-Law model provides a flexible framework for capturing the non-monotonic and nonlinear evolution of surface area during acid-rock dissolution by adjusting the exponent n. This flexibility enhances its ability to respond to variations in mineral dissolution intensity, enabling a more realistic representation of the selective formation and development of dissolution channels. Additionally, the Power-Law model is often coupled with the Kozeny-Carman model to effectively simulate the initiation and propagation of wormholes.

Dispersion tensor model. In porous media, solute transport is governed not only by molecular diffusion but also by fluid flow and the structure of the porous medium. Due to the anisotropic nature of dispersion, its intensity differs across directions and needs to be represented using a tensor. The tortuosity of the porous structure tends to slow down the diffusion rate, while hydrodynamic dispersion stretches the solute band along the flow direction during solute transport. In the simulation of porous media dissolution using the PorousMedia4Foam module, t both mechanisms are captured through the archiesLaw model [30] in this study, which employs a single effective diffusion tensor D_i^* to describe their combined effects.

$$D_i^* = \varphi^n D_i I \tag{7}$$

Where φ is the porosity, D_i is the molecular transport, I is the unit tensor. The archiesLaw model is a way to directly map porosity changes to permeability variations, thereby reflecting how the dissolution process progressively enhances local fluid channels and promotes the formation of wormholes. Furthermore, the model enables fast computation of the nonlinear relationship between porosity and permeability, making it convenient to integrate into reactive transport coupled numerical simulation. Additionally, the model supports heterogeneous structural evolution, which is beneficial for capturing wormhole competition and channel selection mechanisms.

Geometry and initial conditions

The simulation region consists of a two-dimensional rectangular porous media block. The aspect ratio is chosen based on literature [31], with a length of 0.2 m and a width of 0.05 m. The initial porosity is uniformly set to 0.3. A hexahedral structured grid is employed, generated and refined using the snappyHexMesh tool. A porous free flow region [32, 33] with homogeneous and structurally inhomogeneous medium are used here to study the effect of acid injection rate on carbonate dissolution and the formation of wormholes, as shown in Fig. 1. The inhomogeneous structure consists of nine uniformly distributed individual holes. Each hole maintains a square shape with a side length of 2 cm, and each fracture measures 0.5 cm in length. The interiors of the holes are set to be fully developed with a porosity of 0.9. Four constant acid injection rates are applied to investigate their influence on wormhole development: 2.5E-6 cm·s^{-1}, 2.5E-5 cm·s^{-1}, 2.5E-4 $cm \cdot s^{-1}$, and 2.5E-3 $cm \cdot s^{-1}$. A Dirichlet boundary condition is imposed at the inlet (left boundary), maintaining constant velocity and reactant concentration. At the outlet (right boundary), a zero pressure (Neumann) condition is used. No-flow (impermeable) boundary conditions are applied at the top and bottom boundaries. The acid-rock reaction follows a first-order rate law dependent on local surface area and reactant concentration. Dissolution proceeds over time, dynamically altering the local porosity field and hence the permeability according to the Kozeny-Carman law.

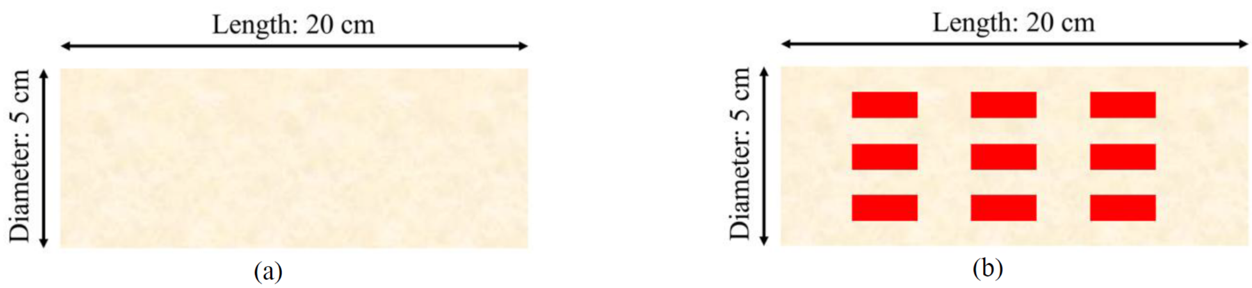

Figure 1—Diagram of the free flow region sample with complex media development: (a) Homogeneity; (b) Heterogeneity

Results and discussions

Fig. 2 investigates the wormholes evolution during acid-induced carbonate dissolution under four distinct injection rates: 2.5E-6 $cm \cdot s^{-1}$, 2.5E-5 $cm \cdot s^{-1}$, 2.5E-4 $cm \cdot s^{-1}$, and 2.5E-3 $cm \cdot s^{-1}$. The red regions represent fully dissolved zones with porosity approaching 1 (i.e., fully dissolved channels), while the blue regions represent undissolved or weakly altered rock matrix. At the lowest velocity of 2.5E-6 $cm \cdot s^{-1}$, acid is rapidly consumed near the inlet due to limited convective transport, resulting in a compact and localized dissolution area without distinct wormhole development. As the injection rate increases to 2.5E-5 $cm \cdot s^{-1}$, a single wormhole channel begins to form and propagate deeper into the medium, suggesting the onset of preferential dissolution paths driven by moderate mass transport. When acid injection rate is 2.5E-4 $cm \cdot s^{-1}$, the wormhole structure becomes significantly more complex and branched, with multiple channels extending through the entire domain, indicating an optimal regime for wormhole formation where acid transport and reaction kinetics are effectively balanced. In contrast, at the highest velocity of 2.5E-3 $cm \cdot s^{-1}$, the acid front advances too quickly to allow sufficient rock dissolution, leading to a broad and diffuse dissolution pattern with poorly defined channels, characteristic of the convection-dominated regime. These results highlight the strong dependence of wormhole morphology on injection velocity. Wormhole formation is favored at intermediate velocities where convective transport is sufficient to deliver acid deeper into the formation without overwhelming the dissolution process. This aligns with theoretical predictions based on the interplay between the Damköhler and Péclet numbers [34, 35]. In sum, the acid-carbonate dissolution remains localized with limited penetration at low injection rate, while intermediate injection rates lead to the formation of highly elongated wormhole structures due to the balance between convection and reaction. At high velocity, dissolution becomes diffuse and inefficient, indicating a transition to convection-dominated transport. These results demonstrate the critical role of injection rate in controlling wormhole morphology.

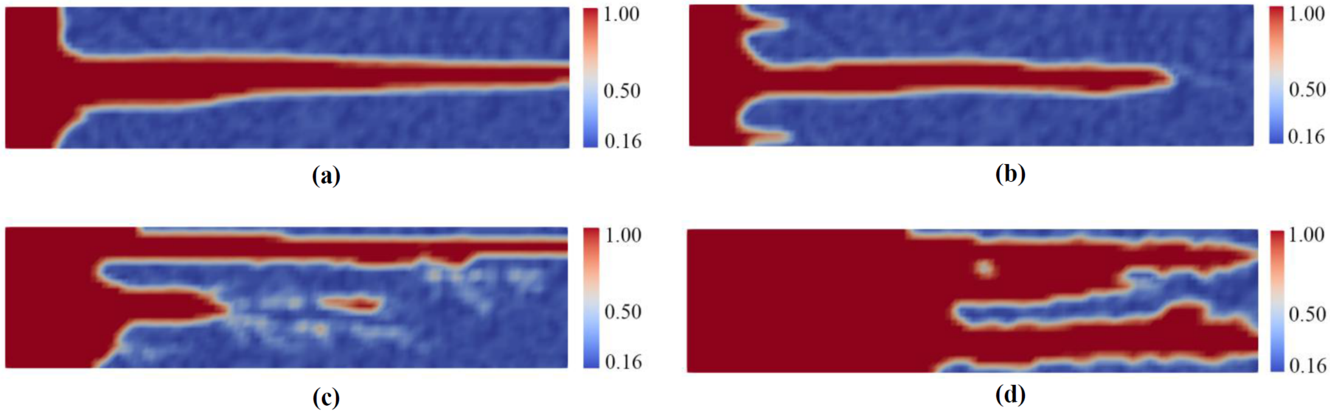

Figure 2—Effect of the different acid injection rates on the acid dissolution process and the formation of wormholes in a homogeneous porous region, where (a) 2.5E-6 cm·s^{-1}; (b) 2.5E-5 cm·s^{-1}; (c) 2.5E-4 cm·s^{-1}, and (d) 2.5E-3 cm·s^{-1}.

Fig. 3 presents the wormhole formation after acid injection under varying acid injection rates in a structurally heterogeneous carbonate medium. From top-left to bottom-right, the acid injection rates are 2.5E-6 cm·s^{-1}, 2.5E-5 cm·s^{-1}, 2.5E-4 cm·s^{-1}, and 2.5E-3 cm·s^{-1}, respectively. As the same of Fig. 2, the red color denotes fully dissolved regions, while blue indicates undissolved matrix. At the lowest flow rate 2.5E-6 cm·s^{-1}, the dissolution pattern begins to display periodic lateral protrusions perpendicular to the main flow direction, a sharp contrast to the compact front observed in the homogeneous case with the same acid injection rate. This indicates that even at reaction-limited conditions, structural irregularities such as pore-scale heterogeneities or mineral inhomogeneities, can perturb acid propagation paths, inducing heterogeneous acid dissolution of carbonates and asymmetric and multi-directional wormhole initiation.

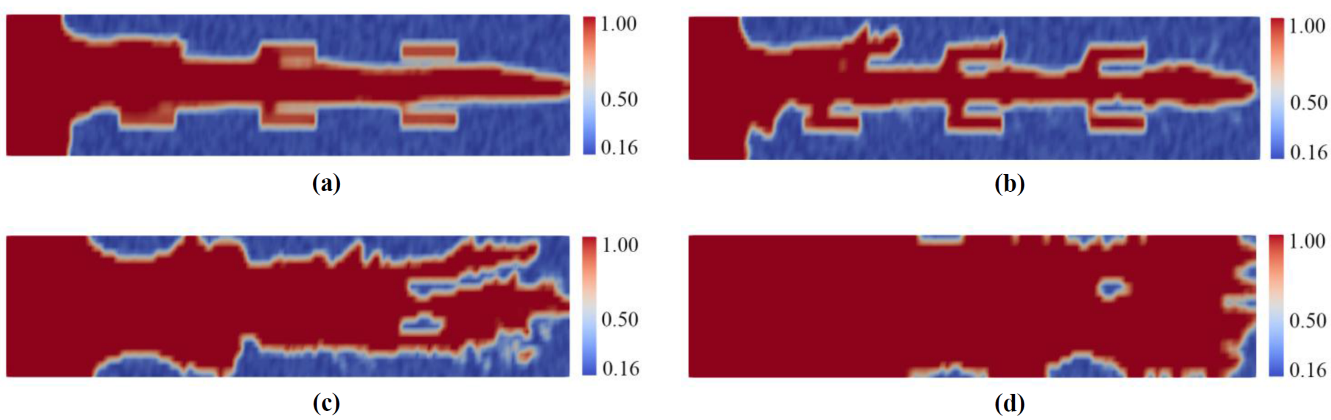

Figure 3—Effect of the different acid injection rates on the acid dissolution process and the formation of wormholes in a heterogeneous porous region, where (a) 2.5E-6 cm·s^{-1}; (b) 2.5E-5 cm·s^{-1}; (c) 2.5E-4 cm·s^{-1}, and (d) 2.5E-3 cm·s^{-1}.

When the acid injection rate increases to 2.5E-5 cm·s^{-1}, the number and extent of lateral wormhole branches increase significantly. These side channels are irregular and tend to cluster, suggesting that structural disorder promotes competitive wormhole growth and destabilizes the main channel. Unlike the homogeneous case where a single dominant channel forms, the heterogeneous setting results in multiple co-evolving dissolution fronts. At 2.5E-4 cm·s^{-1} of acid injection rate, dissolution becomes highly irregular and fractal-like. Here multiple wormholes propagate in parallel, forming a sponge-like morphology. The channels are less straight and more tortuous compared to the homogeneous zone. This indicates that under optimal Péclet and Damköhler conditions, heterogeneity amplifies the nonlinearity of dissolution and enhances morphological complexity. At the highest acid injection rate 2.5E-3 cm·s^{-1}, acid-carbonate dissolution becomes nearly uniform throughout the domain. However, small undissolved patches remain, surrounded by high-porosity zones. Compared to the homogeneous case, where dissolution was broad but more evenly distributed, here the residual heterogeneity leads to discrete acid bypassing and inefficient utilization, highlighting the detrimental effect of fast convection in disordered systems. Overall, the comparison reveals that structural heterogeneity dramatically alters the wormhole morphology, increasing branching, asymmetry, and competition among dissolution fronts. These results suggest that wormhole optimization in practical applications requires accounting for natural rock heterogeneity, which significantly impacts dissolution efficiency and channel development.

Conclusions

This study systematically investigates the influence of acid injection rate and geological heterogeneity on the dissolution behavior of carbonate rocks. By comparing homogeneous and heterogeneous porous media under four flow conditions, we reveal distinct wormhole formation mechanisms and morphological differences. The key findings are summarized as follows:

1. Injection velocity governs wormhole formation regimes. Here moderate acid injection rates produce optimal, focused wormholes, while low or high injection rates lead to inefficient dissolution.
2. Geological heterogeneity amplifies morphological complexity. In heterogeneous media, structural irregularities strongly perturb the dissolution front, leading to the formation of asymmetric, branched, and spatially irregular wormhole networks. Even at the same acid injection rate, the dissolution morphology differs significantly from that in homogeneous media.
3. Wormhole competition and acid bypassing emerge under high injection rate in heterogeneous media. At high injection rates in heterogeneous region, acid tends to bypass less permeable zones, leaving undissolved pockets.
4. Design of acidizing and carbon storage strategies must account for rock heterogeneity. Simplified homogeneous models may significantly overestimate treatment efficiency or underestimate breakthrough time, so accurate prediction of reactive flow and channel morphology requires incorporating spatial variability.

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References

1. Alameedy U, Fatah A, Abbas A K, et al Matrix acidizing in carbonate rocks and the impact on geomechanical properties: A review. *Fuel*, 2023, **349**: 128586.
2. Luo E, Fan Z, Hu Y, et al A dual-porosity dual-permeability model for acid gas injection process evaluation in hydrogen-carbonate reservoirs. *International Journal of Hydrogen Energy*, 2019, **44**(46): 25169–25179.
3. Szymczak P, Kwiatkowski K, Jarosinskí M, et al Wormhole formation during acidizing of calcite-cemented fractures in gas-bearing shales. *SPE Journal*, 2019, **24**(02): 795–810.
4. Cai P, Yan H, Wang H, et al Wormholing in fractured porous media: Nonlocal reactive flow modelling based on peridynamic theory. *Computers and Geotechnics*, 2024, **176**: 106755.
5. Akanni O O, Nasr-El-Din H A, Gusain D. A computational Navier-Stokes fluid-dynamics-simulation study of wormhole propagation in carbonate-matrix acidizing and analysis of factors influencing the dissolution process. *SPE Journal*, 2017, **22**(06): 2049–2066.
6. Morse J W, Arvidson R S. The dissolution kinetics of major sedimentary carbonate minerals. *Earth-Science Reviews*, 2002, **58**(1-2): 51–84.

7. Soulaine C, Roman S, Kovscek A, et al. Mineral dissolution and wormholing from a pore-scale perspective. *Journal of Fluid Mechanics*, 2017, **827**: 457–483.
8. Kalia N, Balakotaiah V. Modeling and analysis of wormhole formation in reactive dissolution of carbonate rocks. *Chemical Engineering Science*, 2007, **62**(4): 919–928.
9. Soulaine C, Pavuluri S, Claret F, et al porousMedia4Foam: Multi-scale open-source platform for hydro-geochemical simulations with OpenFOAM. *Environmental Modelling & Software*, 2021, **145**: 105199.
10. Maes J, Soulaine C, Menke H P. Improved volume-of-solid formulations for micro-continuum simulation of mineral dissolution at the pore-scale. *Frontiers in Earth Science*, 2022, **10**: 917931.
11. Dong K, Zhu D, Hill A D. Mechanism of wormholing and its optimal conditions: A fundamental explanation. *Journal of Petroleum Science and Engineering*, 2018, **169**: 126–134.
12. Zhang L, Wang H, Zhou F, et al Numerical simulation of wormhole propagation with foamed-viscoelastic-surfactant acid in carbonate acidizing. *Processes*, 2023, **11**(6): 1839.
13. Zhang, L., Wang, X., Li, Y. Wormhole propagation regimes in heterogeneous carbonates: A PorousMedia4Foam investigation. *Journal of Porous Media*, 2025, **18**(2): 101–115.
14. Chang T, Jiang Y, Li Y, et al Study on the effect of natural fractures and temperature on the wormhole morphology formed by two-phase acidizing in carbonate rocks. *Physics of Fluids*, 2024, **36**(8).
15. Doe, J., Zhang, L. Impact of acid injection rate on wormhole patterns: PorousMedia4Foam simulations and experimental validation. *Journal of Petroleum Science & Engineering*, 2023, **220**: 110789.
16. Zhang, L., Smith, K. Reactive transport modeling of calcite dissolution in carbonate rocks using OpenFOAM. *Journal of Geophysical Research: Earth Surface*, 2023, **128**(4): 1–15.
17. Chen, X., Al-Muntasheri, G. Convolutional neural network-assisted prediction of wormhole propagation paths in carbonate acidizing. *SPE Journal*, 2024, **29**(3): 1456–1472.
18. Wang, Y., Li, H. Generative adversarial networks for rapid wormhole pattern synthesis in reactive transport simulations. *Computational Geosciences*, 2025, **29**(2): 321–335.
19. Soulaine C, Pavuluri S, Claret F, et al porousMedia4Foam: Multi-scale open-source platform for hydro-geochemical simulations with OpenFOAM. *Environmental Modelling & Software*, 2021, **145**: 105199.
20. Pavuluri S, Tournassat C, Claret F, et al Reactive transport modeling with a coupled openfoam-phreeqc platform. *Transport in Porous Media*, 2022, **145**(2): 475–504.
21. Goyeau B, Lhuillier D, Gobin D, et al Momentum transport at a fluid-porous interface. *International Journal of Heat and Mass Transfer*, 2003, **46**(21): 4071–4081.
22. Soulaine C, Tchelepi H A. Micro-continuum approach for pore-scale simulation of subsurface processes. *Transport in porous media*, 2016, **113**: 431–456.
23. Brinkman H C. A calculation of the viscous force exerted by a flowing fluid on a dense swarm of particles. *Flow, Turbulence and Combustion*, 1949, **1**(1): 27–34.
24. You J, Lee K J. Pore-scale study to analyze the impacts of porous media heterogeneity on mineral dissolution and acid transport using Darcy-Brinkmann-Stokes Method. *Transport in Porous Media*, 2021, **137**: 575–602.
25. Molins S, Soulaine C, Prasianakis N I, et al Simulation of mineral dissolution at the pore scale with evolving fluid-solid interfaces: Review of approaches and benchmark problem set. *Computational Geosciences*, 2021, **25**: 1285–1318.
26. Kozeny J. *Über kapillare Leitung des Wasser sim Boden*. Sitzungsbe-richte: Royal Academy of Science, Vienna, 1927.

27. You J, Lee K J. Analyzing the dynamics of mineral dissolution during acid fracturing by pore-scale modeling of acid-rock interaction. *Spe Journal*, 2021, **26**(02): 639–652.
28. Soulaine C, Roman S, Kovscek A, Tchelepi H A. Mineral dissolution and wormholing from a pore-scale perspective. *J. Fluid Mech*. 2017, **827**: 457–483.
29. Noiriel C, Luquot L, Mad'e B, Raimbault L, Gouze P, van der Lee J. Changes in reactive surface area during limestone dissolution: an experimental and modelling study. *Chem. Geol*. 2009, **265** (1-2): 160–170.
30. Wang H, Revil A. Surface conduction model for fractal porous media. *Geophysical Research Letters*, 2020, **47**(10): e2020GL087553.
31. Al-Harthy S, Nasr-El-Din H A. Numerical modeling and studies of the acid stimulation process in dolomite carbonate rocks. *Journal of Petroleum Science and Engineering*, 2024, **221**: 111203.
32. Jia C, Yao J, Sepehrnoori K. *Acid Stimulation Simulation Theory and Application in Carbonate Rocks*. 2025.
33. Huang Z Q, Liu K, Cui S Z, et al Numerical simulation of reactive flow in fractured vuggy carbonate reservoirs considering hydro-mechanical-chemical coupling effects. *SPE Journal*, 2025, **30**(02): 853–874.
34. Fredd C N, Fogler H S. Influence of transport and reaction on wormhole formation in porous media. *AIChE journal*, 1998, **44**(9): 1933–1949.
35. Ming L I U, Zhang S, Jianye M O U. Fractal nature of acid-etched wormholes and the influence of acid type on wormholes. *Petroleum Exploration and Development*, 2012, **39**(5): 630–635.